University of Management & Technology
Department of Artificial Intelligence
School of Systems and Technology
Fall Semester 2025

Instructor's Name:	**Mr. Syed Hamed Raza**	**Subject:**	Discrete Structures
Course Code	CC-141	**Assignment #:**	01
Total Marks	30	**Due Date:**	**14-11-2025**
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Question 01 **(2 Marks)**

Using set laws and Venn diagrams, verify whether

$$A - (A - B) = A \cap B$$

is always true for any two sets A B
Represent both sides of the equation with an example.

Question 02: **(2 Marks)**

Let $S = \{\emptyset, \{a\}, \{a, b\}\}$.

1. **Write all elements of $P(S)$.**
2. **Find $|\ P(S)\ |$and $|\ P(P(S))\ |$.**
3. **State the general rule connecting a set and its power set.**

Question 03: **(2 Marks)**

Let the universal set represent Pakistani students:

$$U = \{\text{Ali,Sara,Hassan,Ayesha,Bilal,Fatima,Usman,Nida}\}$$

Define:
$A = \{$Ali,Sara,Ayesha,Usman$\}$ students who submitted the assignment.
$B = \{$Sara,Ayesha,Bilal,Fatima,Usman$\}$ students who passed the course.

Write the elements of each:

1. $A \cap B$
2. $B - A$
3. $A \cup B$

4. $A^c \cap B$
   Describe what each set represents in one line.

Question 04: **(2 Marks)**
In a survey of 50 students:

- 20 use Python,
- 15 use R,
- 18 use SQL,
- 8 use both Python and R,
- 6 use both Python and SQL,
- 5 use both R and SQL,
- 3 use all three languages.

Find the number of students who:

1. Use only Python
2. Use only R
3. Use all three languages
4. Use none of the three

Draw a labeled Venn diagram.

Question 05: **(2 Marks)**
Given the following sets:

$$A = \{1, 2, 3, 4\}, B = \{2, 4, 6, 8\}, C = \{4, 8, 12, 16\}$$

Determine whether each of the following statements is true or false:

1. $A \subseteq B$
2. $A \cap C = \{4\}$
3. $(A \cup B) - C = \{1, 2, 3, 6\}$
4. $A^c \cap B = B - A$

Question 06: **(2 Marks)**
Define $a_n = (3n - 2)(4n + 5)$.

1. Write the first five terms.

2. Express $S_n = a_1 + a_2 + \cdots + a_n$.

3. Evaluate S_3.

Question 07: **(2 Marks)**

For $a_n = (2n+1) \cdot 3^{n-1}$:

1. **Write first six terms.**
2. **Find total of first n terms.**
3. **Describe how the sequence grows.**

Question 08: **(2 Marks)**

Given $a_n = 5a_{n-1} - 6a_{n-2}$, $a_0 = 2$, $a_1 = 5$:

1. **Compute a_2, a_3, a_4, a_5.**
2. **Write general form of a_n.**
3. **Describe whether the growth is linear or exponential.**

Question 09: **(2 Marks)**

Sequence: $1, 3, 4, 7, 11, 18, 29, 47, 76, 123$.

10. **Identify the rule generating this sequence.**
11. **Write next three terms.**
12. **Classify as linear or non-linear.**

Question 10: **(2 Marks)**

Evaluate the sum of the first n terms of the series:

$$2 + 6 + 12 + 20 + 30 + \cdots$$

Write the formula for the n^{th}term and the total S_n.
(Hint: Each term can be written as $n(n+1)$).

Question 11: **(2 Marks)**

Consider the series:

$$1 \cdot 5 + 5 \cdot 11 + 9 \cdot 17 + 13 \cdot 23 + \cdots$$

1. **Find the general term of the series.**
2. **Derive the formula for S_n(sum to $n$terms).**
3. **Write simplified expression for $n = 4$.**

Question 12: (2 Marks)
Universal set of programming students:

$$U = \{\text{Ali,Sara,Hassan,Ayesha,Bilal,Fatima,Usman,Nida}\}$$

$A = \{$Ali,Sara,Hassan,Ayesha$\}$ enrolled in AI
$B = \{$Ayesha,Bilal,Fatima,Usman$\}$ enrolled in DBS

Compute and list:

1. $A \cup B$
2. $A \cap B$
3. $A - B$
4. $B - A$
5. $A \Delta B$
6. A^c

Question 13: (2 Marks)
The financial sequence is defined as

$$R_n = 0.5R_{n-1} + 0.2R_{n-2} + 0.3n, R_0 = 1, R_1 = 2$$

Find R_2, R_3, R_4.
Discuss whether the sequence stabilizes or increases indefinitely.
Interpret "stability" in a real-world context.

Question 14: (2 Marks)
Suppose you have:
$A = \{1, 2, 3\}, B = \{2, 3, 4\}, C = \{3, 4, 5\}$.

- **Compute $A \times B$and $B \times C$.**
- **Find the number of ordered triples in $A \times B \times C$.**
- **Write the first five elements of $A \times B \times C$.**

- **State whether $(\boldsymbol{A} \times \boldsymbol{B}) \times \boldsymbol{C} = \boldsymbol{A} \times (\boldsymbol{B} \times \boldsymbol{C})$.**

Question 15: **(2 Marks)**

For the geometric sequence $\mathbf{2, 6, 18, 54, \ldots}$:

- **Write the common ratio.**
- **Find the 7th term.**